MEGHA SHANTHRAJ

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PROFESSIONAL SUMMARY

AI-focused Senior Software Engineering Lead with 20+ years of experience designing and delivering distributed systems, now building production-grade AI applications at Gauntlet AI. Expertise spans LLM orchestration, multi-agent systems, RAG pipelines, agentic tool-use architectures, and real-time AI interfaces — layered on a deep foundation in AWS serverless, microservices, and enterprise-scale system design. Track record of leading cross-functional teams, driving architectural decisions, and shipping mission-critical integrations processing millions of transactions. Certified Java professional (SCJP, SCWCD, OCA).

CORE COMPETENCIES

AI & LLM Systems	Multi-Agent Orchestration, LLM-as-Judge Evaluation, RAG Pipelines (LlamaIndex, Weaviate, Voyage Code 2), Agentic Architectures (LangGraph, ReACT), Claude/GPT-4o Function Calling, MCP Server Design, Prompt Engineering, Real-Time Voice AI (LiveKit, Deepgram, Simli)
System Design	Distributed Microservices, Event-Driven Architecture, API Gateway Design (APIGEE), Serverless Patterns, WebSocket Real-Time Systems, Iframe Sandboxing & Trust/Safety, CI/CD Pipeline Architecture, Observability & Monitoring
Languages	Python, TypeScript, Java, Node.js, JavaScript, SQL, PL/SQL
Cloud & DevOps	AWS (Lambda, Step Functions, DynamoDB, S3, API Gateway, CodePipeline, SAM, Cognito, Route 53, Redshift, CloudFormation, X-Ray), Docker, Railway, Vercel, Fly.io, Firebase, Hetzner VPS, SonarQube, CI/CD
Frameworks	FastAPI, Next.js, React, Angular, NestJS, Spring Boot, Spring Security, Hibernate, LangChain, LlamaIndex, LangGraph
Data & Infra	PostgreSQL, MongoDB, DynamoDB, Redis, SQLite, Supabase, SQL Server, Oracle, Weaviate, Socket.IO, ActiveMQ, JMS
Tools & Practices	Git, Claude Code, Claude Desktop (Code + Cowork), Cursor, Jira, Confluence, Splunk, AppDynamics, Postman, Agile/Scrum, TDD, BDD

PROFESSIONAL EXPERIENCE

Gauntlet AI — Feb 2026 – Present

AI Software Engineer — Intensive Training Program

Austin, TX | Building production-grade AI applications across multi-agent systems, RAG, real-time AI, and agentic architectures

Tech: Python, TypeScript, FastAPI, Next.js, React, Angular, NestJS, Claude API, GPT-4o, LangChain, LangGraph, LlamaIndex, Weaviate, Supabase, LiveKit, Deepgram, Simli, Socket.IO, PostgreSQL, Docker, MCP Servers, Cloudflare Workers

- **Agent Fortio — AI Portfolio Intelligence Agent:** Built a conversational AI agent for real-time investment portfolio analysis on Ghostfolio. Implemented dual-interface delivery (CLI + MCP server with 11 specialized tools) enabling native integration with Claude Desktop. Engineered a LangGraph-based ReACT loop with 5-stage verification pipeline (cross-validation, confidence scoring, hallucination detection). Tech: Python, FastAPI, LangGraph, NestJS, Prisma, Claude function calling.
- **LegacyLens — RAG-Powered Codebase Intelligence:** Designed a retrieval-augmented generation system making 250K+ lines of LAPACK Fortran queryable via natural language. Implemented hierarchical chunking with Voyage Code 2 embeddings and Weaviate hybrid search (vector + BM25). Built RouterQueryEngine for multi-strategy retrieval, achieving >70% precision@5 across 25 golden evaluations. Deployed on Railway with Next.js frontend. Tech: Python, FastAPI, LlamaIndex, Weaviate, Next.js.
- **BuddyTutor — Real-Time Socratic AI Video Tutoring Avatar:** Architected a sub-second-latency AI video avatar tutor for 6th–12th graders using a concurrent streaming pipeline (STT → LLM → TTS → Avatar). Implemented Socratic method prompt engineering with interrupt handling, VAD-based turn detection, and targeted rewrite loops. Tech: Python, LiveKit Agents, Deepgram, Groq, Cartesia TTS, Simli, Next.js.
- **Ad System — Autonomous Ad Copy Generation with LLM-as-Judge:** Built a multi-model ad generation pipeline (GPT-4o-mini for generation, Claude Sonnet for 5-dimension evaluation) with iterative feedback loops. Implemented quality threshold enforcement

(7.0/10 minimum), targeted dimension regeneration, calibration gating, and cost tracking ($0.014/publishable ad). Generated 100+ evaluated ads with full scoring traces. Tech: Python, FastAPI, OpenAI API, Anthropic API, Langfuse, PostgreSQL.
• **ChatBridge — AI Chat Platform with Plugin Architecture:** Designed production-grade architecture for a K-12 AI chatbot platform with third-party app integration via iframe sandboxing and postMessage protocol. Engineered trust & safety layer with app verification pipelines, CSP enforcement, and schema-validated tool registration. Tech: Next.js 14, Supabase, Socket.IO, Claude API.
• **AgentRelay — AI Agent Orchestration Platform:** Built a full-stack platform for creating, configuring, and orchestrating collaborative AI agents with real tool execution (file I/O, shell commands, web search, HTTP requests). Implemented a visual workflow builder (React Flow) with DAG execution, conditional routing, feedback loops, and shared workspaces enabling multi-agent collaboration. Features include an app launcher that auto-detects and runs agent-generated code, a guardrails engine (rate limiting, cost caps, tool blocking), Telegram bot integration, and real-time WebSocket monitoring. Deployed on Hetzner VPS with Docker. Tech: Python, FastAPI, Next.js 14, React Flow, PostgreSQL (JSONB), Redis Pub/Sub, SQLAlchemy 2.0, Claude/GPT-4o function calling, Docker, Caddy.
• **MCP Server — Satellite Imagery Integration:** Built a remote MCP server on Cloudflare Workers wrapping SkyFi's satellite imagery API, enabling natural language satellite image search, ordering, and area monitoring across 150+ satellites. Integrated OpenStreetMap geocoding, implemented quote-then-order workflow with human-in-the-loop purchase confirmation. Tech: TypeScript, Cloudflare Workers, MCP Protocol, SSE transport.
• **ZapPulse — Event Ingestion & Delivery API:** Built a RESTful event ingestion, persistence, and delivery platform with three delivery models: pull-based inbox polling, webhook push with exponential backoff retry (5 attempts), and real-time SSE streaming. Implemented API key authentication, sliding-window rate limiting, idempotency keys, dead letter queue, and ULID-based cursor pagination. Includes a React Explorer UI with dashboard charts, event composer, and live feed. Deployed on Railway. Tech: Python, FastAPI, SQLite (WAL), React, Vite, Tailwind CSS, SSE, Pydantic v2.
• **AI Coach — Healthcare Exercise Accountability Partner:** Built an AI-powered accountability coach for home exercise programs using LangGraph for phase-based conversation routing (Pending → Onboarding → Active → Re-Engaging → Dormant). Implemented hybrid safety classifier (regex + LLM) blocking clinical advice, consent gate on every interaction, 5 LangChain tools (set_goal, alert_clinician, set_reminder), and disengagement handling with exponential backoff follow-ups. 62 automated tests with zero API key dependency. Tech: Python, FastAPI, LangGraph, LangChain, Claude, SQLite, LangSmith, Docker.
• **Additional projects**: Collab Canvas (collaborative workspace), Leader Link (leadership communication tool). Details available upon request.
- Developed a reusable AI-first project inception workflow: five-phase pipeline (Digest → Research → Interview → Decisions → Spec) with gate enforcement, NFR checklists, and Claude Code skill integration for repeatable, high-quality project scaffolding.

University of California, Los Angeles (UCLA) Jul 2010 – August 2025

Senior Software Engineering Lead (Consultant)

Ascend 2.0 | Enterprise Platform (Apr 2013 – Present)

Tech: Python, Node.js, AWS (Lambda, Step Functions, DynamoDB, S3, API Gateway, CodePipeline, SAM, SES, Cognito, ACM), APIGEE, JWT, GitHub, GoAnywhere FX, Redshift

• Architected and delivered 15+ serverless microservices using AWS Lambda, Step Functions, and DynamoDB, eliminating dedicated server provisioning while improving scalability and fault tolerance for university-wide integrations.
• Designed and deployed secure REST APIs through APIGEE proxy gateway with Cognito OAuth authentication, enabling controlled access for 10+ downstream campus systems processing thousands of daily transactions.
• Built automated CI/CD pipelines using AWS SAM templates, CodePipeline, and CodeBuild, cutting deployment time from hours to minutes and enabling zero-downtime releases across dev, staging, and production environments.
• Engineered end-to-end file integration workflows using GoAnywhere Director for PGP encryption/decryption, SFTP transfers, and S3 synchronization, processing sensitive financial and payroll data for 45,000+ university employees.
• Led cross-functional requirements gathering with 8+ stakeholder teams and boundary systems, translating business needs into technical integration specifications and driving alignment across distributed teams.
- Managed and mentored 5+ offshore developers, conducting systematic design and code reviews, establishing coding standards, and ensuring on-time delivery across multiple concurrent workstreams.

Senior Software Developer (Consultant)

Time Reporting System (UCPATH), Effort Reporting, Financial & Payroll Applications (Jul 2010 – Mar 2023)

Tech: Java (JDK 9), Spring, Spring Security, Hibernate, REST, React.js, Node.js, jQuery, SQL Server, Git, Quartz Scheduler, AWS

• Designed and implemented the core profile loading batch process, migrating data from PeopleSoft operational data stores to the Time Reporting System's SQL Server database, handling 45,000+ employee records with validation and error recovery.
• Developed and maintained the Time Reporting System, an in-house time tracking application built for UCLA and deployed across 80+ departments at UCLA, UC Irvine, UC Merced, and UCOP, serving thousands of employees and supervisors for daily time entry and approval workflows.
• Built timesheet creation and management modules with complex hours calculation logic supporting multiple union contracts, overtime rules, compensatory time accrual, shift differentials, and holiday pay — ensuring compliance with UC-wide labor policies and union-specific collective bargaining agreements.
• Led system analysis and requirements gathering for timesheet submission and approval workflows, producing technical specifications and process flowcharts adopted as the reference architecture for UC-wide rollout.
• Configured and hardened Linux server infrastructure (Apache, SSL, Shibboleth SSO, Tomcat), automating deployment through shell scripts and JAMS scheduler, reducing manual deployment effort by 80%.

- Coordinated multi-campus rollout across UCLA, UC Irvine, UC Merced, and UCOP, enabling rapid iteration through close collaboration with functional and technical teams.

Fidelity Business Services Mar 2007 – May 2010

Track Lead — Technical and Functional

Insight Trading Engine & MaintApps for Net Holdings/Escrow | Bangalore, India

Tech: Java, Spring 2.5 (IoC, AOP), Spring MVC, Tiles, Hibernate, JavaScript, JMS, WSDL, Oracle, Tomcat

• Led a 10-member cross-functional team (BAs, developers, QA) through full sprint lifecycle — planning release scope, facilitating daily standups, resolving blockers, and driving on-time delivery of trading engine modules.

• Architected and implemented core trading engine functionality for the Insight platform, processing real-time equity and fund transactions with Spring-managed transaction boundaries and Hibernate persistence.

- Received "On the Spot" award for delivering critical trading engine features ahead of schedule, and "All Hands Recognition" award for outstanding track leadership and proactive risk escalation.

Oracle Financial Services (formerly i-Flex Solutions) Mar 2006 – Mar 2007

FlexCube Developer

FLEXCUBE Universal Banking Platform | Bangalore, India

Tech: Oracle PL/SQL, PL/SQL Developer

- Developed and optimized PL/SQL modules for the FLEXCUBE universal banking platform, building stored procedures and packages supporting core banking operations for enterprise clients across Asia and the Middle East.

EDUCATION

Masters in Business Administration — Finance

Bachelor of Engineering — Information Science & Engineering

CERTIFICATIONS

• Sun Certified Java Programmer (SCJP)

• Sun Certified Web Component Developer (SCWCD)

- Oracle Certified Associate (OCA)

AWARDS

• "On the Spot" Award — Fidelity Business Services — Delivering core trading engine features ahead of schedule.

• "All Hands Recognition" Award — Fidelity Business Services — Exceptional track leadership, timely delivery, and proactive issue escalation.